

It is recommended that you spend about 60 minutes on this question.

For
Examiner's
Use

- 1 In this experiment you will investigate the oscillations of a loaded multi-spring system.

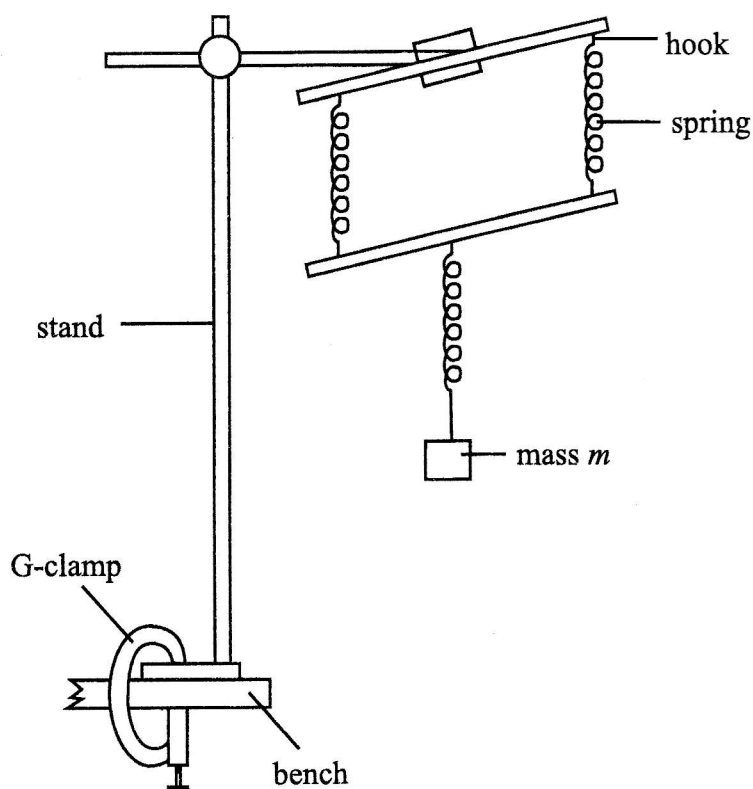


Fig. 1.1

- (a) (i) Set up the apparatus as shown in Fig. 1.1 with the initial mass, m of 100 g.
- (ii) Slightly displace the mass, m so that it performs oscillations in a vertical plane.
- (iii) Make and record measurements to determine the period, T of oscillation of the system.
- (iv) Increase the mass, m for m in the range $100 \text{ g} \leq m \leq 600 \text{ g}$ and repeat (i) to (iii) to obtain six sets of readings.

In the table of results, include values of m , $\log m$ and $\log T$.

- (v) Justify the number of significant figures used for $\log T$.

- (b) It is suggested that T and m are related by the equation,

$$T = \beta m^{\alpha}$$

where α and β are constants.

- (i) Plot a graph of $\log T$ (y-axis) against $\log m$ (x-axis).
- (ii) Use the graph to determine the values of α and β .

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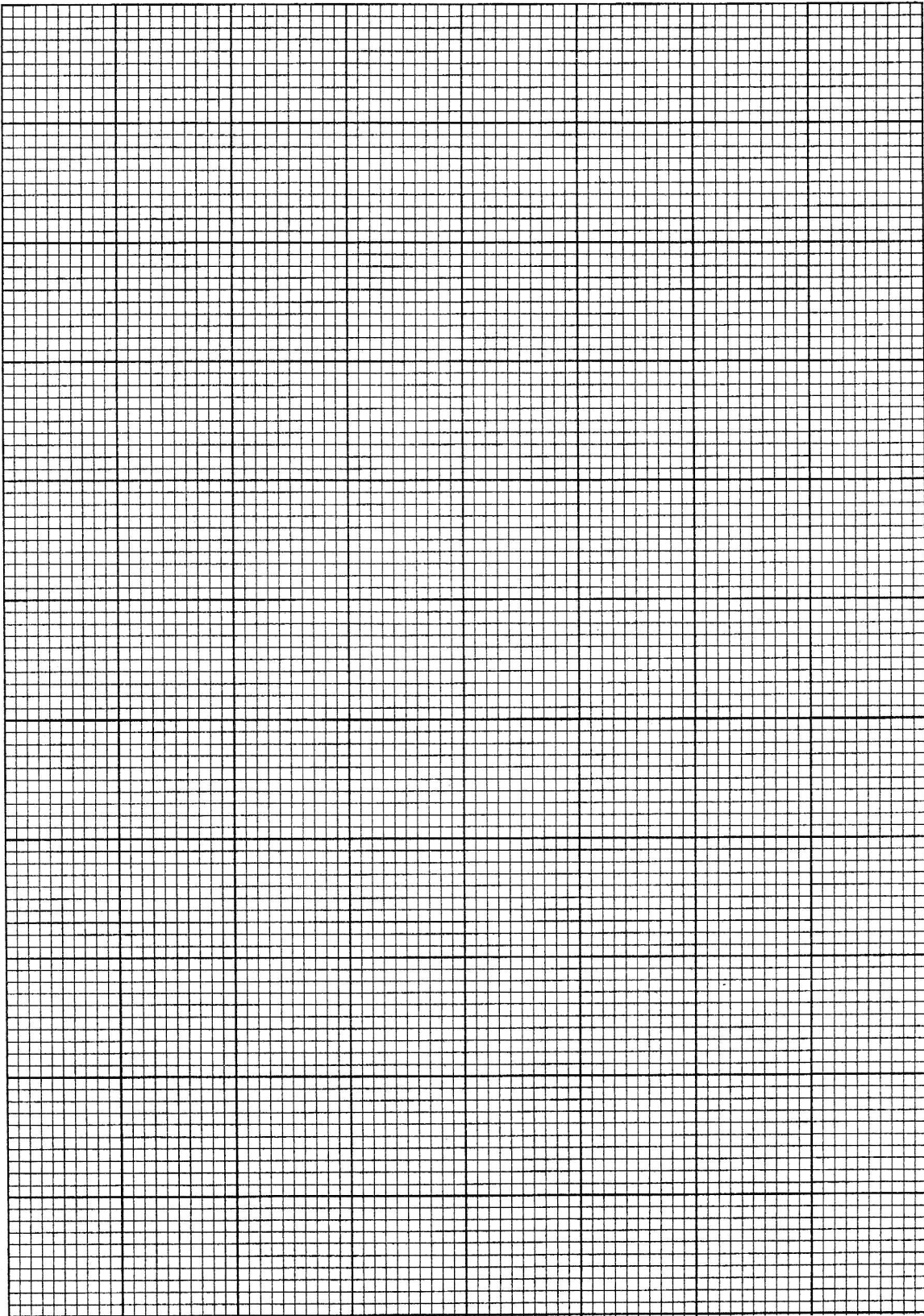
Measurements and calculations

M

R

A

G



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- 2 In this experiment you will investigate the discharging of a capacitor.

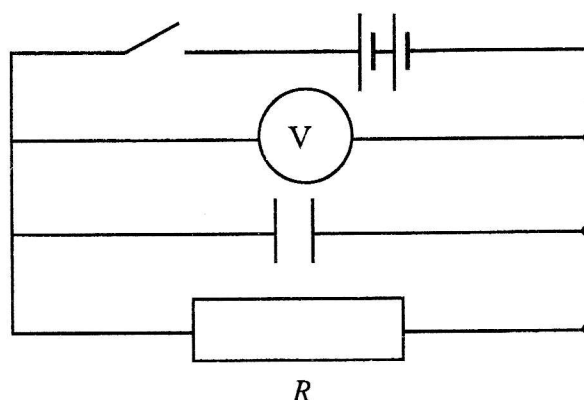


Fig. 2.1

- (a) (i) Set up the circuit shown in Fig. 2.1 with $R = 100\ \Omega$.
 (ii) Close the switch and record the voltmeter reading, V_0 .

Voltmeter reading $V_0 =$ _____

- (iii) Calculate and record V where $V = 0,37 \times V_0$

$V =$ _____

- (iv) Open the switch and simultaneously start the stopwatch to measure the time, t , for the voltmeter reading to fall to V . Record the value of t and that of R in a table.
 (v) Change the value of R and repeat (iv) to obtain six sets of readings of R and t for R in the range $100\ \Omega \leq R \leq 350\ \Omega$.

- (b) (i) Plot a graph of t (y-axis) against R (x-axis).
- (ii) Determine the capacitance of the capacitor given that it is equal to the gradient of the graph.
- (iii) Use the graph to determine the value of t when $R = 2\,000\ \Omega$.
- (c) Use the apparatus of **Fig. 2.1**, to measure and record t using a $2\,000\ \Omega$ resistor provided.

$t =$ _____

- (d) The resistance of the voltmeter is $5\,000\ \Omega$.

Compare and comment on the values of t obtained in (b)(iii) and (c).

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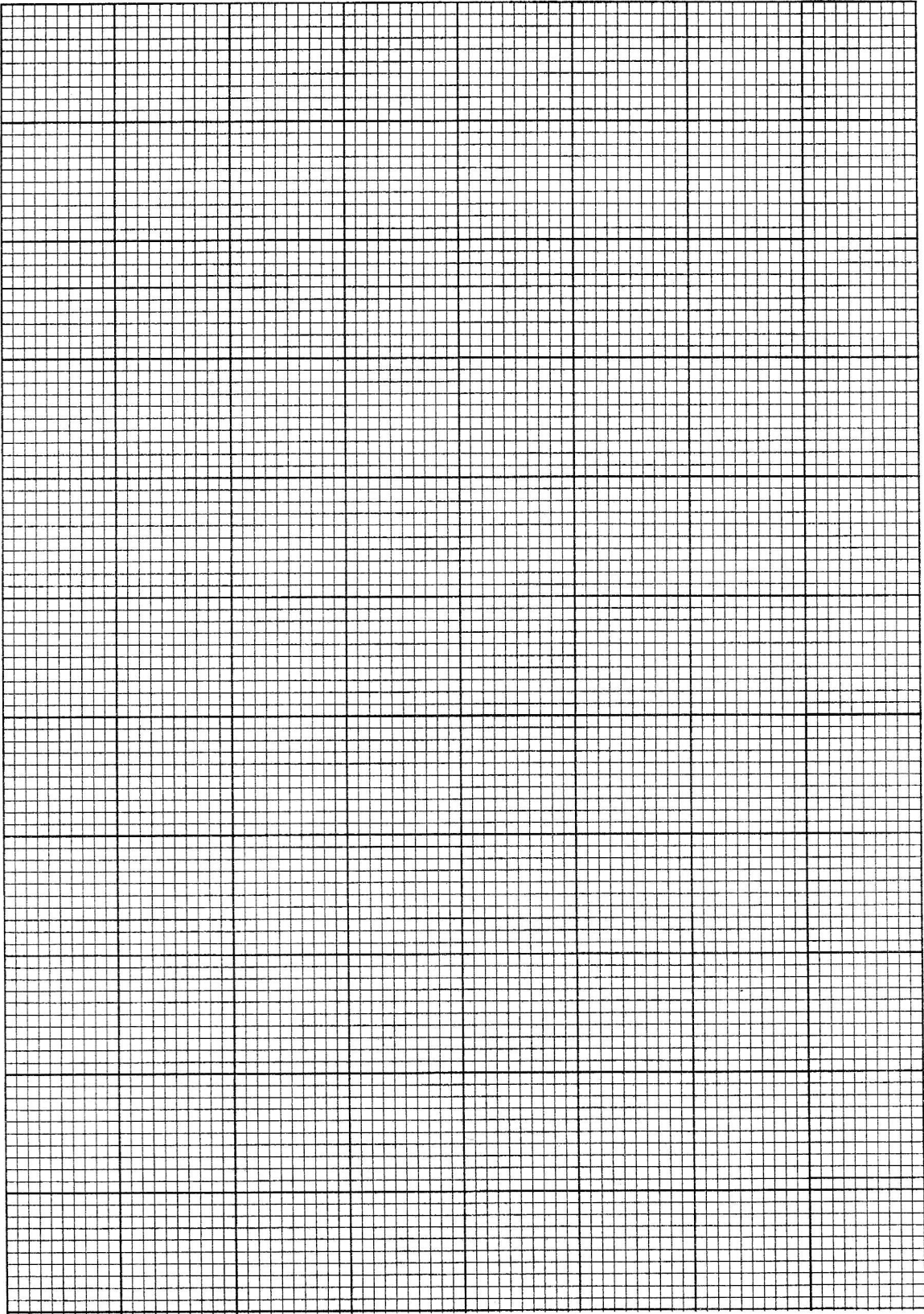
Measurements and calculations

M

R

A

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- 3** The heating effect of a coil in oil depends on the following factors:

material of the coil

number of turns of the coil

the supply voltage

Design a laboratory experiment to show how the listed factors affect the heating effect of the coil in oil.

In your design report, pay particular attention to:

control of variables

procedure to be followed

precautions

good design features.

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This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.
